

# **JNTU Kakinada**

## **MID-TERM EXAMINATION ? 2025**

Analog Electronic Circuits ? Mid-Term Examination, 2025

Subject Code: EC302 Semester: 3 Regulation: R23

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Diode Circuits)**

1. Define Diode Circuits. Explain its significance in Analog Electronic Circuits with a suitable example. [12 Marks]
2. Describe the working principle of Diode Circuits with necessary equations. [12 Marks]
3. List the advantages and limitations of Diode Circuits. Explain any two in detail. [12 Marks]

### **SECTION B (Unit 2: BJT Amplifiers)**

4. Compare and contrast the key concepts involved in BJT Amplifiers. [12 Marks]
5. Derive the expression related to BJT Amplifiers and explain each term. [12 Marks]
6. Discuss the engineering applications of BJT Amplifiers. [12 Marks]

### **SECTION C (Unit 3: FET Amplifiers)**

7. Compare and contrast the key concepts involved in FET Amplifiers. [12 Marks]
8. Derive the expression related to FET Amplifiers and explain each term. [12 Marks]
9. Describe the working principle of FET Amplifiers with necessary equations. [8 Marks]

### **SECTION D (Unit 4: Feedback Amplifiers)**

10. Compare and contrast the key concepts involved in Feedback Amplifiers. [12 Marks]
11. Compare and contrast the key concepts involved in Feedback Amplifiers. [8 Marks]
12. List the advantages and limitations of Feedback Amplifiers. Explain any two in detail. [12 Marks]

### **SECTION E (Unit 5: Oscillators)**

13. Derive the expression related to Oscillators and explain each term. [12 Marks]
14. Discuss the engineering applications of Oscillators. [12 Marks]
15. Derive the expression related to Oscillators and explain each term. [12 Marks]

\*\*\* End of Question Paper \*\*\*